2025-11-18

Chapter 4

1. Give five different path names for the file /etc/passwd. (Hint: Think about the directory entries ‘‘.’’ and ‘‘..’’.)
6. Some operating systems provide a system call rename to give a file a new name. Is there any difference at all between using this call to rename a file and just copying the file to a new file with the new name, followed by deleting the old one?
10. Consider the directory tree of Fig. 4-8. If /usr/jim is the working directory, what is the absolute path name for the file whose relative path name is ../ast/x?
12. Describe the effects of a corrupted data block for a given file for: (a) contiguous, (b) linked, and (c) indexed (or table based).
13. One way to use contiguous allocation of the disk and not suffer from holes is to compact the disk every time a file is removed. Since all files are contiguous, copying a file requires a seek and rotational delay to read the file, followed by the transfer at full speed. Writing the file back requires the same work. Assuming a seek time of 5 msec, a rotational delay of 4 msec, a transfer rate of 80 MB/sec, and an average file size of 8 KB, how long does it take to read a file into main memory and then write it back to the disk at a new location? Using these numbers, how long would it take to compact half of a 16-GB disk?
16. Consider the i-node shown in Fig. 4-13. If it contains 10 direct addresses and these were 8 bytes each and all disk blocks were 1024 KB, what would the largest possible file be?
19. It has been suggested that efficiency could be improved and disk space saved by storing the data of a short file within the i-node. For the i-node of Fig. 4-13, how many bytes of data could be stored inside the i-node?
23. Consider a 4-TB disk that uses 4-KB blocks and the free-list method. How many block addresses can be stored in one block?
25. The beginning of a free-space bitmap looks like this after the disk partition is first formatted: 1000 0000 0000 0000 (the first block is used by the root directory). The system always searches for free blocks starting at the lowest-numbered block, so after writing file A, which uses six blocks, the bitmap looks like this: 1111 1110 0000 0000. Show the bitmap after each of the following additional actions:
    (a) File B is written, using five blocks.
    (b) File A is deleted.
    (c) File C is written, using eight blocks.
    (d) File B is deleted.
36. Consider the idea behind Fig. 4-21, but now for a disk with a mean seek time of 6 msec, a rotational rate of 15,000 rpm, and 1,048,576 bytes per track. What are the data rates for block sizes of 1 KB, 2 KB, and 4 KB, respectively?
38. Given a disk-block size of 4 KB and block-pointer address value of 4 bytes, what is the largest file size (in bytes) that can be accessed using 10 direct addresses and one indirect block?